



**Verified Carbon
Standard**

GROUPED HYDRO POWER PROJECT BY EKI ENERGY SERVICES LIMITED



Document Prepared by EKI Energy Services Limited

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The project serves the purpose of producing clean electrical energy in a sustainable manner. The project activity involves implementation of 2 turbines of 9.51 MW Mai Beni Hydropower Project (MBHP) and 6.2 MW Lower Jogmai Khola Hydropower Project (LJKHP) which utilize potential energy available for power generation. Therefore, no fossil fuels are involved for power generation in the project activity. The proposed project activity will also reduce the anthropogenic emissions of greenhouse gases into the atmosphere by avoiding business as usual operation of existing fossil fuels-based power plants to produce equivalent amount of electricity.

The details of the 9.51 MW Mai Beni Hydropower Project (MBHP) are mentioned below –

Mai Beni Hydropower Project is a run-of-river (ROR) project located in Ilam District, Province 1 of Nepal. The project will have installed capacity of 9.51MW.

Mai Beni Hydropower Project is located within Mai-Jogmai Rural Municipality and Ilam Municipality, Ilam District, State-1 in the Eastern Region of Nepal. The project site is about 527 km east of Kathmandu. The headworks site is located near the border of Soyang and Namsaling. The powerhouse is located 900m downstream of Mai khola and Jogmai khola confluence.

The details of the 6.2 MW Lower Jogmai Khola Hydropower Project (LJKHP) are mentioned below–

The project activity is a run of the river (ROR) scheme at Jogami River of Ilam District, Mechi Zone of Eastern Development Region, Nepal. Project site is accessible by Ilam-Namsaling motorable road from Tilkenimode to Powerhouse Site which is about 6 km. The same road is reaching to surge tank which is about 7.25 km from Tilkenimode. The powerhouse can also be reached by motorable road about 3.5 km from Silpur which is opposite bank of the powerhouse. Project has constructed access road from powerhouse to headworks.

The electricity generated is supplied to Integrated Nepal Power System (INPS) through 4 km long, 33 kV Single Circuit Transmission line from the project to the 132/33kV Godak sub-station of NEA.

The location of the project

The project location for Project Activity Instances is in Nepal and details are provided in Section 1.12 of this document. The project location details, KML files for Nepal country are submitted along with Joint PDMR.

A brief description of the scenario existing prior to the implementation of the project

The baseline scenario would be either continued use of fossil fuel dominated grid connected electricity generation or fossil fuel used in power plant or carbon intensive electricity.

An estimate of annual average and total GHG emission reductions and removals

During registration of project activity, first project activity instances are included. The average annual GHG emission reduction from the first project activity instance is 22,737 tCO₂e.

Project Activity instance:

The details of the first/current project activity instance are as below-

S.I No.	Customer	Installed Capacity (MW)	Date of Commissioning
1	EKI Energy Services Limited	9.51	01-August-2021
2	EKI Energy Services Limited	6.20	01-October-2021
	Total	15.71	

The annual estimated emission reductions over crediting period are 22,737 tCO₂ /annum.

<u>Audit Type</u>	<u>Period</u>	<u>Program</u>	<u>VVB Name</u>	<u>Number of years</u>
Validation/ Verification	<u>01-August-2021-- 31-December-2022)</u>			
<u>Total</u>				

1.2 Sectoral Scope and Project Type

The project activity falls under consideration of the grouped project activity. The project activity instances as part of the grouped project will have the following parameters.

Sectoral Scope : 01 - Energy industries (renewable / non-renewable sources)
 Project Type : I - Renewable Energy Projects
 Project Category : Each project activity instances will apply only small-scale hydro project
 Methodology : AMS-I.D.: Grid connected renewable electricity generation – Version 18.0

The project activity under consideration is a grouped project activity.

1.3 Project Eligibility

The project activity involves installation and generation of electricity using renewable energy resources i.e., by using the hydro power (renewable sources) replacing electricity supply from a

fossil-fuel dominated electricity, thus leads to reductions of anthropogenic GHG emissions from atmosphere. Hence the project activity is eligible Sectoral scope 1 i.e., energy industries (renewable/ non-renewable sources) under the scope of the VCS Program.

As the electricity produced by all the project activity instance involved in the proposed project is primarily intended to be supplied to the grid, 100% of the electricity generated by the hydro power units is to be accounted for emission reduction, even though the project is grid-connected.

1.4 Project Design

- ☐ The project includes a single location or installation only
- ☐ The project includes multiple locations or project activity instances, but is not being developed as a grouped project
- ☒ The project is a grouped project

Eligibility Criteria

The project is a grouped project and following are the eligibility criteria for inclusion of new project activity instances into the grouped project activity:

- 1) Applicability Conditions: The project activity instances shall meet applicability conditions for applicable methodology AMS I.D version 18.0 as defined in section 3.2.
- 2) Geographical Area: The project activity instances to be included in the grouped project activity will be activities involving renewable electricity generation power plants connection using hydro power located within Nepal.
- 3) Baseline scenario: All Project Activity Instances shall meet the baseline definition as defined in respective methodology and as explained in section 3.4.
- 4) Technology type: The project activity instances to be included in the grouped project activity will be Greenfield activities involving renewable electricity generation power plants using hydro power.
- 5) Additionality: The project activity instances to be added as part of the grouped project will meet additionality criteria as set out in the methodology.
- 6) Start Date: The start date of each project activity instance under the grouped project should not be prior to the start date of the grouped project. The start date of each project activity instance will be determined through documentary evidence.
- 7) Conditions that avoid double counting of emission reductions like unique identifications of project and claiming emission reduction only under one GHG program.
- 8) The Grouped Project specific requirements stipulated by the Entity responsible for coordinating and managing grouped project for conducting local stakeholder consultations and environmental impact assessment (EIA), as applicable.

- 9) The project activity using hydro power project will supply electricity to grid or to users through grid network.

The following table demonstrates how the initial project activity instances being included in the grouped project activity fulfill above mentioned eligibility conditions:

Sr. No	Eligibility Criteria	Project Activity instances eligibility
1	Applicability Conditions: The project activity instances shall meet applicability conditions for applicable methodology (AMS I.D version 18.0) as defined in section 3.2.	All project activity instances meet applicability conditions of methodology (AMS I.D Version 18.0 (project activity instances having capacity less than 15 MW), hence this eligibility criteria are fulfilled. Please refer section 3.2 of this document for methodology applicability criteria.
2	Geographical Area: The project activity instances to be included in the grouped project activity will be activities involving Renewable Electricity Generation and grid using hydropower plant located within Nepal.	All project activity instances being included in the grouped project are located within geographic boundaries of Nepal. Hence this condition is fulfilled. Please refer commissioning certificate for Geographical area of project activity.
3	Baseline scenario: All Project Activity Instances shall meet the baseline definition as defined in respective methodology and as explained in section 2.4.	All initial project activity instances have same baseline as per methodology as detailed in subsequent sections. Hence this eligibility criterion is fulfilled. Please refer section 3.4 of this document for baseline scenario.
4	Technology type: The project activity instances to be included in the grouped project activity will be Greenfield activities involving Renewable Electricity Generation and grid using hydropower. The electricity should be supplied to grid.	All project activity instances are hydro power plants supplying electricity for grid. All are green field projects. Hence this condition is fulfilled.

5	Additionality: The project activity instances to be added as part of the grouped project will meet additionality criteria as set out in respective methodologies - for Small Scale Project instances the methodology applicable will be AMS-I.D.	The additionality for the all-project activity instances included in the grouped project has been demonstrated in section 3.5 of Joint PDMR.
6	Start Date: The start date of each project activity instance under the grouped project should not be prior to the start date of the grouped project. The start date of each project activity instance will be determined through documentary evidence.	The start date of grouped project is considered as the earliest commissioning date of project activity instance and start date of all other project activity instances is after this date. Please refer commissioning certificate for start date of grouped project activity.
7	Conditions that avoid double counting of emission reductions like unique identifications of project and claiming emission reduction only under one GHG program.	The existing and new project activity instances are identified by latitude and longitude and location of the plant. Declaration has been provided and these project activity instances are not participating in any other GHG program. Hence this condition is fulfilled.
8	The Grouped Project specific requirements stipulated by the Entity responsible for coordinating and managing grouped project for conducting local stakeholder consultations and environmental impact assessment (EIA), as applicable.	Local stakeholder consultation has been conducted as a part of grouped project activity. There is no requirement for carrying out EIA for these projects in Nepal, hence this condition is fulfilled.
9	The project activity using hydropower project will supply electricity to grid.	The existing and new project activity instances are hydro power plants supplying electricity to grid. Hence this condition is fulfilled.

As per clause 3.5.7 of VCS Standard Version 4.1, Non-grouped projects with multiple project activity instances shall not exceed any capacity limits to which a project activity is subject. Thus, capacity limits criteria are applicable for non-grouped project activity. The proposed project activity is grouped project activity, thus capacity limit.

As per clause 3.5.10 of VCS Standard Version 4, Grouped project activity allows multiple project activities where meth applied to project allows it. Each project activity instance has total power generation capacity less than 15 MW.

“As with non-grouped projects, grouped projects may incorporate multiple project activities (see Section 3.5.1 – 3.5.3 for more information on multiple project activities). Where a

grouped project includes multiple project activities, the project description shall designate which project activities may occur in each geographic area”.

As per para 3.5.14 of VCS Standard V4.1, each project activity instances are not exceeding the capacity limit of 15 MW. The project activity instance involves multiple project activity instances and it is not cluster of project activity instances. Also, each project activity instances does not exceeding the 1% of capacity limit of 15 MW. Thus, there is no any requirement of division of clusters of project activity instances and no requirement to demonstrate that each project activity instances are not within one kilometer of each other. If any project activity instance is bundled project, then that project activity instance capacity is within capacity limit of 15 MW.

1.5 Project Proponent

Organization name	Urja Developers Private Limited
Contact person	Bhanu Bhakta Pokharel
Title	Director
Address	Krishna Galli-3, Pulchowk, Lalitpur, Nepal
Telephone	977-9851059177
Email	bhanu.pokharel@urjadeveloper.com

Organization name	EKI Energy Services Limited
Contact person	Manish Dabkara
Title	Managing Director & Chief Executive Officer
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Telephone	+91 99075 34900
Email	manish@enkingint.org and registry@enkingint.org

1.6 Other Entities Involved in the Project

Organization name	EKI Energy Services Limited
Role in the project	Project Consultant

Contact person	Mr. Ankit Sethiya
Title	Deputy Manager – Operations
Address	Plot No 48, Scheme 78, Vijay Nagar Part- II, Indore 452010, Madhya Pradesh, India
Telephone	0731 428 9086
Email	ankit.sethiya@enkingint.org

1.7 Ownership

The renewable energy technology installed at site of each of the project activities as part of the proposed project activity are under legal ownership of Urja Developers Private Limited. EKI Energy Services Limited is authorized PP by all owners of hydro power plant. The commissioning certificate for project activity are the supporting documents to demonstrate the project ownership.

This demonstrates the right of use according to clause 3.11.1 (3) of VCS Standard (v4.0) – “a project ownership arising by virtue of a statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals”

1.8 Project Start Date

Start date of project activity is 01- August-2021, which is the date of commissioning of the earliest commissioned project involved in the grouped project activity. Though first project activity is yet to be commission, PP is willing to include the project activity instances during registration having commissioning dates from 01- August-2021 onward and as per deadline of 2 years of validation completion from date of commissioning.

1.9 Project Crediting Period

Crediting Period Start date: 01- August -2021

Crediting Period End date: 31- July -2028

The grouped project activity adopts renewable crediting period of 7 years period which can be renewed for maximum 2 times. This start date of crediting period will vary based on actual earliest commissioned date within the project activity instances of grouped project activity.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

☐ <20,000 tCO₂e/year

☒ 20,000 – 100,000 tCO₂e/year

☐ 100,001 – 1,000,000 tCO₂e/year

☐ >1,000,000 tCO₂e/year

Project Scale	
Project	✓
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
Year 1	22,737
Year 2	22,737
Year 3	22,737
Year 4	22,737
Year 5	22,737
Year 6	22,737
Year 7	22,737
Total estimated ERs	159,162
Total number of crediting years	07
Average annual ERs	22,737

1.11 Description of the Project Activity

The project involves installation of Mai Beni Hydropower Project and Lower Jogmai Khola Hydropower Project.

1. Mai Beni Hydropower Project is a run-of-river (ROR) project located in Illam District, Province 1 of Nepal. The project will have installed capacity of 9.51MW. A power purchase

agreement has also been signed between Samling Power Company and Nepal Electricity Authority, the Government of Nepal undertaking company. The intake site is located near the border of Soyang and Namsaling. The powerhouse is located just downstream of Mai khola and Jogmai khola confluence. Overall project layout is proposed within a stretch of 5-6 km with all components located at the right bank of Mai Khola.

2. Lower Jogmai Khola Hydropower Project is a Run-of-River project, it is essential to conduct long-term hydrological investigation as well as to estimate the low flow and flood flow. As mentioned earlier, Jogmai is an ungauged River. Therefore, data from other River gauge stations has been taken to derive the long-term flow of Jogmai Khola at the intake site.

Installed capacity of the project is optimized at 6.2 MW. This scheme comprises of 20m wide diversion weir with weir crest level at 649 masl, which will divert the design flow into the intake located at the left side of the river. Weir will be built with a layer of boulder lining. An Under sluice having 2m width has been provided before the intake. Sediment settling system of the plant is located at the distance of about 200m from the intake. It has two chambers of size 62m x 6.05m x 6m (L x B x H). The scheme comprises of a 2870m long headrace pipe having diameter of 1.8m. A 55m long surge pipe of 3.0m diameter is designed to arrest the surge in the scheme. Powerhouse is connected by 417m long penstock having 1.8m diameter. Two Horizontal-axis Francis turbines each of 3.10MW capacity will be installed in the powerhouse.

As per para 3.5.14 of VCS Standard V.4.0, each project activity instance of grouped project activity is less than 15 MW capacity. Thus, small scale methodology AMS I.D is applicable for proposed grouped project activity.

As per the applicable methodology, a Greenfield power plant is defined as “a new renewable energy power plant that is constructed and operated at a site where no renewable energy power plant was operated prior to the implementation of the project activity”.

The Project activity is a new facility (Greenfield) and the electricity generated by the Project will be exported to the national Grid. The Project will therefore displace an equivalent amount of electricity which would have otherwise been generated by fossil fuel dominant electricity grid. The Project Proponent plans to avail the VCS benefits for the Project.

1.12 Project Location

The details of the project locations are mentioned in the table below:

S.I No.	Customer	Premise address where the unit is installed	Installed Capacity (MW)
1	EKI Energy Services Limited	Ilam District Province 1, Nepal	9.51
2	EKI Energy Services Limited	Ilam District Province 1, Nepal	6.20

This is a grouped project activity and the Nepal is the project activity. The KML file shared with the DOE.



A KML file have been submitted in order to indicate the project location and geographic boundaries.

1.13 Conditions Prior to Project Initiation

The project is a Greenfield hydro power project and does not involve generation of GHG emissions for the purpose of their subsequent reduction, removal or destruction. Prior to the initiation of the project activity, the equivalent amount of electricity would have been drawn from grid connected or new power plants, in national Grid. The grid is predominantly coal based and therefore is a major source of carbon dioxide emissions in India. The main emission in the pre project scenario is the power plants connected to the national Grid, and main GHG involved is CO₂.

As per methodology, the baseline scenario would be either continued use of fossil fuel dominated grid connected electricity generation or fossil fuel used in power plant or carbon intensive electricity.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The Project has received necessary approvals for development and commissioning for each PV Plant from the state Nodal agencies and is in compliance to the local laws and regulations.

There is no legal requirement on the choice of a particular technology for power generation.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The proposed grouped project activity has not been registered and is not seeking registration at moment under any other GHG programs.

The initial project activity instances are not participating in any other GHG program.

1.15.2 Projects Rejected by Other GHG Programs

The proposed grouped project activity is not participating in other Environmental credits, other GHG programs and has not been rejected by other GHG programs.

The initial project activity instances are not participating in other Environmental credits, other GHG programs and have not been rejected by other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes

☒ No

The grouped project activity under consideration is not participating in any other Emission trading program and other binding limits.

The all project activity instances are not participating at moment in any other emission trading program and other binding limits. The GHG emission reductions generated during the current monitoring period have not been used for compliance of other mechanism.

1.16.2 Other Forms of Environmental Credit

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes

☒ No

The proposed grouped project activity neither has nor intends to generate any other form of GHG related environmental credit for GHG emission reductions or removals claimed under the VCS Program. The initial project activity instances are neither has nor intends to generate any other

form of GHG related environmental credit for GHG emission reductions or removals claimed under the VCS Program.

1.16.3 Supply Chain (Scope 3) Emissions

Not Applicable

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

Contribution to sustainable development:

Apart from generation of renewable electricity, the grouped project activity and project activity instances proposed to be included under it would contribute to the sustainable development of the region - socially, environmentally and economically. Ministry of Environment and Forests, has stipulated economic, social, environment and technological well-being as the four indicators of sustainable development. The project contributes to sustainable development using the following ways.

Social well-being: The project would help in generating employment opportunities during the construction and operation phases. The project activity will lead to development in infrastructure in the region like development of roads and also may promote business with improved power generation.

Economic well-being: The project is a clean technology investment in the region, which would not have been taken place in the absence of the VCS benefits the project activity will also help to reduce the demand supply gap in the state. The project activity creates local employment generation which helps economic well-being of local people.

Technological well-being: The successful operation of project activity would lead to promotion of hydro based power generation and would encourage other entrepreneurs to participate in similar projects.

Environmental well-being: Hydro being a renewable source of energy, it reduces the dependence on fossil fuels and conserves natural resources which are on the verge of depletion. Due to its zero emission the Project activity also helps in avoiding significant amount of GHG emissions.

1.17.2 Sustainable Development Contributions Activity Monitoring

Designated National Authority (DNA) of the host country (Nepal) has stipulated two indicators for sustainable development in the interim approval guidelines for Clean Development Mechanism (CDM) projects in Nepal. The project proponent believes that the project activity has contributed to sustainable development in terms of the two indicators.

The project supplies clean electricity from the hydro power project to the national Grid, hence displacing the electricity generated from grid connected fossil fuel power plants and thereby avoiding the equivalent Carbon dioxide which is a Green House Gas.

This section will complete during validation and verification stage.

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2	7.2.1: Renewable energy share in the total final energy consumption	Implemented activities to increase	MWh of renewable electricity has been supplied to national grid during the reported period that helps to increase the renewable energy share in the energy mix.	About MWh renewable electricity has been supplied to Indian grid that helps to increase the renewable energy share in the energy mix.
2)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	By supplying MWh clean electricity to national grid, the project avoided release of tCO _{2e} in to the atmosphere during the reporting period.	Prevented the release of tCO _{2e} into the atmosphere.

1.18 Additional Information Relevant to the Project

Leakage Management

Since the project activity instances to be included into the grouped project activity are renewable energy projects based on hydropower, no leakage emissions are considered - in line with paragraph 27 of the small-scale methodology AMS.I.D (version 18.0).

Thus, leakage management plan and implementation of leakage and risk mitigation measures are not applicable for this grouped project activity.

Commercially Sensitive Information

Not applicable. No any commercially sensitive information has been excluded from the public version of the project description. There is no commercially sensitive information.

Further Information

Not Applicable

2 SAFEGUARDS

2.1 No Net Harm

The project activity promotes environmental and socio-economic well-being as it results in zero GHG emissions due to installation and operation of clean, renewable energy technology for electricity generation.

Thus, there are no any significant impacts due to implementation of project activity on air, water, soil quality and ambience are envisaged due to the project activity.

2.2 Local Stakeholder Consultation

The local stakeholder consultation section will be completed during validation stage.

2.3 Environmental Impact

For hydropower projects in Nepal the EIA study has to comply with a range legal requirements of government of Nepal like Constitution of Nepal, 2072, Policy, Guidelines, Acts, Regulations and International Treaties and Conventions such as: Three Years man 2070 (2013), Hydropower Development Policy, 2001, Forest Policy, 2000, Nepal Biodiversity strategy, 2002, Nepal Biodiversity Strategy Implementation man, 2006, National Water strategy (2002) and Nepal Water man 2005, National Environmental Impact Assessment Guidelines, 1993, Department of

Electricity Development Manuals, Forestry Sector ETA Guidelines 1995, Forest Produce, Collection, Sale and Distribution Guideline, 2000, Community Forest Guidelines, 2008, Community Forest Inventory Guidelines, 2005, Environmental Management Guidelines (Road), 1997, Guidelines for the physical Infrastructure Development and Operation in the Protected Areas 2065 (2008), soil and Watershed Conservation Act. 1982, Forest Act, Convention on Biological Diversity, 1992 etc.

2.4 Public Comments

This section will complete after 30 days of public comment period.

2.5 AFOLU-Specific Safeguards

This section is not required as this project activity is a non-AFOLU project activity.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The project activity instances as part of the proposed grouped project activity will be small scale project activity instances with power generation capacity less than or equal to 15 MW as per CDM terminology. Accordingly, below mentioned methodology will be used for each of the project activity instances.

The project activity instance based on its capacity:

The methodology for the project activity instances will be the approved methodology for small- scale CDM project activities. The details of the methodology are as follows:

Methodology: AMS.I.D

Project Type	: Type-I: Renewable Energy Projects
Title	: AMS I. D Grid Connected Renewable Electricity Generation
Version No.	: Version 18.0; EB 81
Reference	: CDM Methodologies ¹

¹ <https://cdm.unfccc.int/methodologies/DB/W3TINZ7KKWCK7L8WTFQOQFQOH4SBK>

3.2 Applicability of Methodology

The small-scale grid connected project activity instances under grouped project activity will meets the applicability conditions of the approved methodology AMS I.D, Version 18.0, Sectoral Scope 1, as described below-

Applicability Conditions	Project Applicability
1. This methodology comprises renewable energy generation units, such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass: (a) Supplying electricity to a national or a regional grid. (b) Supplying electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling.	The project activity is a Hydro Power Project that supply electrical power to the regional electricity grids i.e., National Grid.
2. Illustration of respective situations under which each of the methodology (i.e. AMS-I.D, AMS-I.F and AMS-I.A) applies is included in Table 2²	The 1st & 3rd option of Table 2 of AMS I.D, Version 18, EB 81 is applicable (please refer footnote).
3. This methodology is applicable to project activities that (a) install a new power plant at a site where there was no renewable energy power plant operating prior to the implementation of the project activity (Greenfield plant); (b) involve a capacity addition; (c) involve a retrofit of (an) existing plant(s); or (d) involve a replacement of (an) existing plant(s).	The project activity is installation of new power plant at site where there was no renewable energy power plant operating prior to the implementation of the project activity (Greenfield plant). Hence this condition is applicable.
4. Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: The project activity is implemented in an existing reservoir with no change in the volume of reservoir;	The project activity is a run of river project activity. Hence this condition is not applicable.

2

	Project type	AMS-I.D
1	Project supplies electricity to a national/regional grid	Applicable
2	Project displaces grid electricity consumption (e.g., grid import) and/or captive fossil fuel electricity generation at the user end (excess electricity may be supplied to a grid)	Not Applicable
3	Project supplies electricity to an identified consumer facility via national/regional grid (through a contractual arrangement such as wheeling)	Applicable
4	Project supplies electricity to a mini grid system where in the baseline all generators use exclusively fuel oil and/or diesel fuel	Not Applicable
5	Project supplies electricity to household users (included in the project boundary) located in off grid areas	Not Applicable

- The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the Project Emissions section, is greater than 4 W/m²; - The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the Project Emissions section, is greater than 4 W/m².	
5. If the new unit has both renewable and non-renewable components (e.g., a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	Not applicable, the project activity is the Renewable Energy Project i.e., Hydro Power Project.
6. Combined heat and power (co-generation) systems are not eligible under this category.	Not applicable, the project activity is the Renewable Energy Project i.e., Hydro Power Project
7. In the case of project activities that involve the addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.	The project activity doesn't involve the addition of renewable energy generation units at an existing renewable power generation facility; hence this condition is not applicable.
8. In the case of retrofit or replacement, to qualify as a small-scale project, the total output of the retrofitted or replacement unit shall not exceed the limit of 15 MW.	The project activity doesn't retrofit or replace an existing facility; hence this condition is not applicable.
9. In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid, then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as "AMS-I.C.: Thermal energy production with or without electricity" shall be explored.	The Project activity is a hydro power project and is not a landfill gas, waste gas, waste water treatment and agro-industries projects or recovered methane emissions project. Hence the criteria is not applicable.
10. In case biomass is sourced from dedicated plantations, the applicability criteria in the tool "Project emissions from cultivation of biomass" shall apply.	The Project activity is a hydro power project and is not a biomass project. Hence the criteria is not applicable.

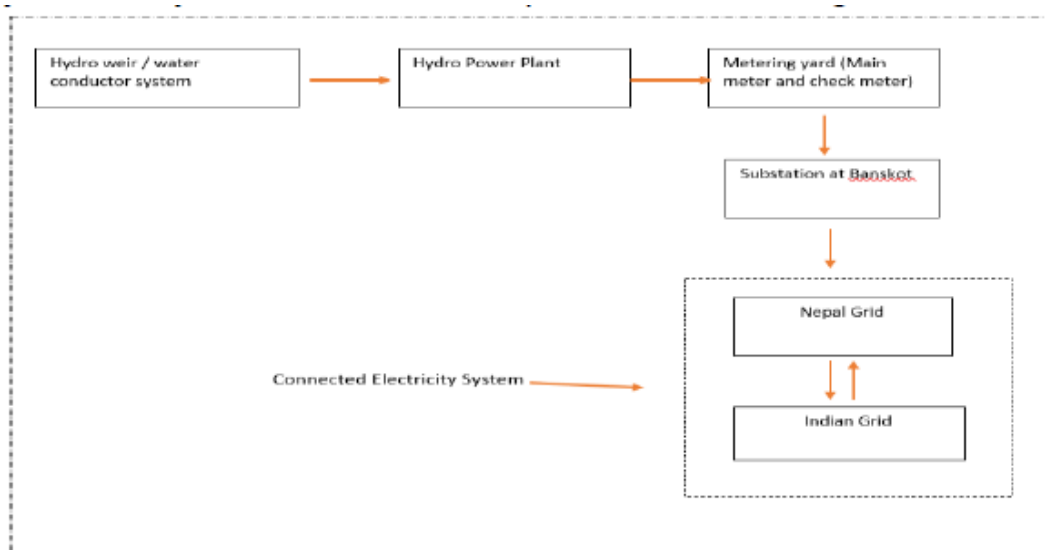
Applicability conditions of “Tool to calculate the emission factor for an electricity system”	
1. This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g., demand-side energy efficiency projects).	This condition is applicable. OM, BM and CM are estimated using the tool under section B.6.1 for calculating baseline emissions.
2. Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off grid power plants. In the latter case, the conditions specified in “Appendix 2: Procedures related to off grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	Since the project activity is grid connected, this condition is applicable and the emission factor has been calculated accordingly.
3. In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The project activity is located in Nepal, a non-Annex I country. Therefore, this criterion is not applicable.
4. Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	The project activity is a grid connected hydro power project and not a hydro power plant. Therefore, this criterion is not applicable.

The project activity qualifies as Type I during every year of the crediting period in accordance with applicable provisions for project activity eligibility as discussed above. Also, the total installed capacity of each project activity instance less than 15 MW threshold limit of small-scale project activity instances and respective methodology will be applicable for that project activity instances.

3.3 Project Boundary

As per AMS-I.D Version 18, EB 81 – “The spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the hydro project power plant is connected to”.

The project boundary is therefore the physical boundary, which includes diversion weir, intake chamber, de-silting chamber, power channel, forebay, headrace tunnel, penstock, powerhouse, tailrace, the transmission system to the evacuation point and auxiliary power consumed by the plant. The below figure illustrates the boundary of project activity. Therefore, the entire national grid and all connected power plants have been considered in the project boundary for the proposed project activity.



Source		Gas	Included?	Justification/Explanation
Baseline	Grid connected electricity generation	CO2	Yes	Main emission source
		CH4	NO	Minor emission source
		N2O	NO	Minor emission source
		Other		
Project	Greenfield Hydro Power Project activity	CO2	NO	No CO2 emissions are emitted from the project
		CH4	NO	Project activity does not emit CH4
		N2O	NO	Project activity does not emit N2o
		Other		

3.4 Baseline Scenario

As per the applicable methodology, a Greenfield power plant is defined as “a new renewable energy power plant that is constructed and operated at a site where no renewable energy power plant was operated prior to the implementation of the project activity”.

Nepal is interconnected with India at various places through 11kV, 33kV, 132kV and 220kV lines. For transfer of bulk power, interconnection between India and Nepal through Dhalkebar (Nepal) - Muzaffarpur (India) 400kV D/C transmission line has been constructed. A total of about 700 MW of power is being supplied to Nepal through these interconnections.

In the regional grid consisting of Nepal and the Indian grid, electricity delivered to the grid by the project would have otherwise been generated by the operation of grid-connected power plants operational by the addition of new generation sources into regional grid.

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as described in section 4.1.

3.5 Additionality

The additionality of the Project shall be demonstrated by applying the following approach, consisting of two components:

- (i) A Legal Requirement Test; and
- (ii) An Additionality Test either based on a Positive List test or a projects-specific additionality test.

(a) Legal Requirement Test

The project is not enforced by law. The project passes the legal requirement test since there are no enforced laws, statutes, regulations, court orders, environmental-mitigation agreements, permitting conditions of other legally-binding mandates requiring its implementation. Since voluntary commitments/agreements within a sector or by an entity do not constitute the legal requirement, the outcome of the legal requirement test is concluded as positive.

(b) Additionality Test

Additionality Assessment

As per the ‘Guidelines on the demonstration of Additionality of small-scale Project Activity’ (version 9.0), a positive list of grid-connected renewable electricity generation technologies is listed that are automatically defined as additional, without further documentation of barriers. The positive list comprises of the following grid-connected renewable electricity generation technologies of installed capacity up to 15 MW.

The Additionality section will be completed during validation stage.

3.6 Methodology Deviations

Not Applicable

4 IMPLEMENTATION STATUS

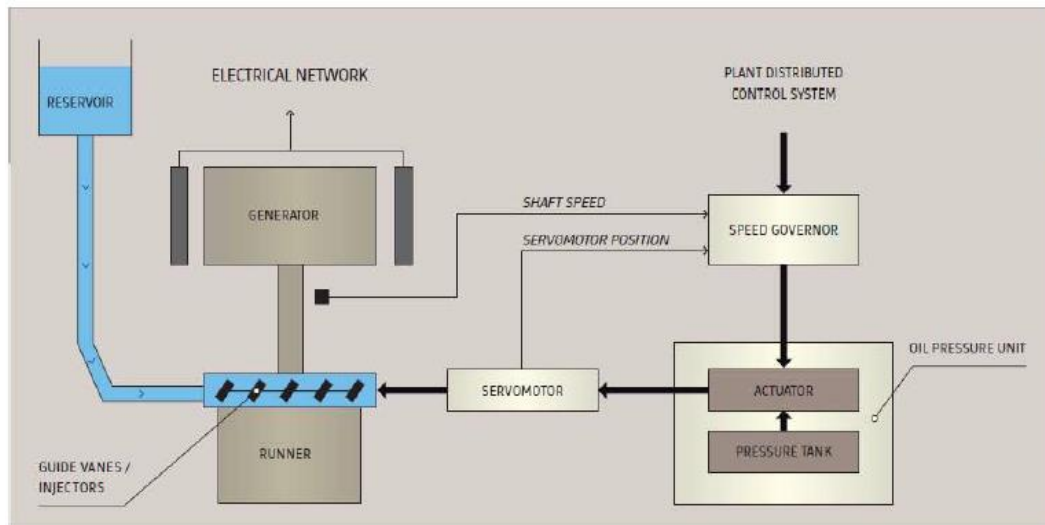
4.1 Implementation Status of the Project Activity

The project activity involves installation and generation of electricity using renewable energy resources i.e., by using the hydro power (renewable sources) replacing electricity supply from a fossil-fuel dominated electricity, thus leads to reductions of anthropogenic GHG emissions from atmosphere.

The electricity generated is supplied to Integrated Nepal Power System (INPS) through 4 km long, 33 kV Single Circuit Transmission line from the project to the 132/33kV Godak sub-station of NEA. The operational life time of project activity is 30 years.

Moreover, the operation of the project activity(s) during this monitoring period, including any information events that may impact GHG emission reduction or removals and monitoring is not applicable. And no breakdown occurred in this plant during the verification period of the project activity.

The main equipment diagram of the hydro power plant:



5 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

The project activity mainly reduces carbon dioxide through substitution of grid electricity generation with fossil fuel fired power plants by renewable electricity. As per para 43 of the methodology AMS I D Version 18, the emission reduction ER_y by the project activity during a given year y is the difference between baseline emissions (BE_y), project emissions (PE_y) and emissions due to leakage (LE_y), as follows:

$$ER_y = BE_y - PE_y - LE_y$$

where:

ER_y = Emission reductions in year y (t CO₂/y)

BE_y = Baseline Emissions in year y (t CO₂/y)

PE_y = Project emissions in year y (t CO₂/y)

LE_y = Leakage emissions in year y (t CO₂/y)

Baseline Emissions:

Baseline emissions in the project activity is calculated as per para-22 of AMS I.D. version-18:

$$BE_y = EG_{PJ,y} \times EF_{grid,y}$$

where:

BE_y = Baseline emissions in year y (t CO₂)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the GCC project activity in project year (MWh)

$EF_{grid,y}$ = CO₂ emission factor for grid connected power generation in year (tCO₂/MWh)

- **Calculation of the CO₂ emission factor of the grid**

As per “Tool to calculate the Emission Factor for an electricity system version-07.0”, following steps has to be applied:

- Step 1: Identify the relevant electricity systems;
- Step 2: Choose whether to include off-grid power plants in the project electricity system (optional);
- Step 3: Select a method to determine the operating margin (OM);
- Step 4: Calculate the operating margin emission factor according to the selected method;
- Step 5: Calculate the build margin (BM) emission factor;
- Step 6: Calculate the combined margin (CM) emission factor

Step 1: Identify the relevant electricity systems

Nepal is interconnected with India at various places through 11kV, 33kV, 132kV and 220kV lines⁹. For transfer of bulk power, interconnection between India and Nepal through Dhalkebar (Nepal) - Muzaffarpur (India) 400kV D/C transmission line has been constructed. A total of about 7008 MW of power is being supplied to Nepal through these interconnections.

Further, 400kV D/C Gorakhpur (India) – Butwal (Nepal) line, 400kV D/C Dhalkebar (Nepal) – Sitamarhi (India) line, 132kV D/C Nanpara, Bihar (India) – Kohalpur (Nepal), stringing of second circuit of 132kV line Kataiya (India) – Kushaha (Nepal) and 132 kV Raxaul (India) – Parwanipur (Nepal) lines have been agreed.

As per Para 15 and 16 of 'Tool to calculate the Emission Factor for an electricity system version-07.0' (Tool 7), Project owner has identified Project electricity system and connected electricity systems as Nepal and Indian+Nepal Grid respectively.

As per para 17 of 'Tool to calculate the Emission Factor for an electricity system version-07.0' (Tool 7),

Project Owner has chosen the 'Option 3': A delineation of the project electricity system defined by more than one independent dispatch areas, e.g., multi-national power pools.

As per para 18 of 'Tool to calculate the Emission Factor for an electricity system version-07.0' (Tool 7): In case of option 3, transmission lines between dispatch areas included in the proposed delineation shall be checked for the existence or non-existence of transmission constraints.

As per para 19 of the 'Tool to calculate the Emission Factor for an electricity system version-07.0' (Tool 7):

There are no transmission constraints if any one of the following criteria is met:

(b) In case of electricity systems with spot markets for electricity: there are differences in electricity prices (without transmission and distribution costs) of less than five per cent between the two electricity systems at least during 90 per cent of the hours of the most recent year for which information is available (at least one year data is required); or

(c) The transmission line(s) is operated at 75 per cent or less of its rated capacity during 90 per cent or more of the hours of the most recent year for which information is available (at least one year data is required) using the algorithm below:

- (i) For every hour of the year check whether the transmission line is operated at 75 per cent or less of its rated capacity;
- (ii) Each hour of the year when the transmission line was operated at 75 per cent or less of its rated capacity should be counted as zero;
- (iii) Each hour of the year when the transmission line was operated at 75 per cent or more of its rated capacity should be counted as one;
- (iv) There is no transmission constraint if the total sum is less than ten per cent of the hours of the year (e.g., 876 for even year and 878 for leap year);
- (v) The algorithm can be illustrated by the following equation:

$$\sum_{1}^{8760} \left[\frac{\text{Hourly power transmission (MWh)}}{[\text{Maximum line's load capacity (MW)}]} > 75\text{per cent} \right] < 876$$

(vi) The maximum line's load capacity should be based on official information (e.g., from the operator of the system);

(d) The transmission capacity of the transmission line(s) that is connecting electricity systems is more than 10 per cent of the installed capacity either of the project electricity system or of the connected electricity system, whichever is smaller.

Project Owner has shown below that there is no transmission constraint according to para 19 (c):
Transmission capacity of the transmission line(s) that is connecting electricity systems (Indian grid to Nepal Grid) = 700 MW

Installed Capacity of the project electricity system (Nepal10) = 1451.34 MW
Installed Capacity of connected electricity system (India) = 3,95,075 MW

Hence, 10 % of the project electricity system is 145.13 MW. And thus, the transmission capacity of the transmission line(s) that is connecting electricity systems (700 MW) is more than 10 per cent of the installed capacity (145.13 MW).

As per Para 25 of 'Tool to calculate the Emission Factor for an electricity system version-07.0' (Tool 7): Project owner has chosen option b as mentioned in tool to calculate the simple operating margin emission rate of the exporting grid (in this project activity Indian grid is the exporting grid). And details have mentioned in Step 3.

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional)

Project participants may choose between the following two options to calculate the operating margin and build margin emission factor:

Option I: Only grid power plants are included in the calculation.

Option II: Both grid power plants and off-grid power plants are included in the calculation.

As the project is exporting electricity to the NEWNE regional grid, thus, the project participants have chosen Option I for the calculation of the operating and build margin emission factor i.e., off grid power plants are not being included in the calculation.

Step 3: Select a method to determine the operating margin (OM)

The calculation of the operating margin emission factor ($EF_{\text{grid,OM,y}}$) is based on one of the following methods:

- (a) Simple OM, or
- (b) Simple adjusted OM, or
- (c) Dispatch data analysis OM, or

The data required to calculate Simple adjusted OM and Dispatch data analysis OM is not possible due to lack of availability of data to project developers. The choice of other two options for calculating operating margin emission factor depends on generation of electricity from low-cost/must-run sources. In the context of the methodology low cost/must run resources typically include hydro, geothermal, wind, low-cost biomass, nuclear and solar generation.

Share of Low cost Must-Run (Hydro/Nuclear) (% of Net generation)

	2015-16	2015-16	2017-18	2018-19	2019-20	2020-21
India (Connected System)	15.1%	15.1%	14.3%	14.5%	17.0%	16.5%

Data Source: Central Electricity Authority (CEA) database Version 17.0, October 2021

The above data clearly shows that the percentage of total grid generation by low-cost/ must-run plants (on the basis of average of five most recent years) for the Indian grid is less than 50 % of the total generation. Thus, the Average OM method cannot be applied, as low cost/must run resources constitute less than 50% of total grid generation.

The Simple OM emission factor is calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (t CO₂/MWh) of all generating power plants serving this system, not including low-cost/must-run power plants/units.

For the simple OM, the emissions factor can be calculated using either of the following data vintages.

a) Ex-ante options: If the ex-ante option is chosen, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emission factor during the crediting period is required.

Or

b) Ex-post option: If the ex-post option is chosen, the emission factor is determined for the year in which the project activity displaces grid electricity, requiring the emissions factor to be updated annually during monitoring.

PP has chosen ex-ante option for calculation of Simple OM emission factor using a 3 years' generation-weighted average, based on the most recent data available at the time of submission of PSF for validation.

OM determined at validation stage will be the same throughout the crediting period. There will be no requirement to monitor & recalculate the emission factor during the crediting period.

Step 4: Calculate the operating margin emission factor ($EF_{grid, OM Simple, y}$) according to the selected method

The operating margin emission factor has been calculated using a 3-year data vintage:

Net Generation in operating Margin (GWh) (Incl. Imports)

	2018-19	2019-20	2020-21
Indian Grid (Connected System)	9,95,957	9,65,009	9,58,218

Simple Operating Margin (t CO₂/MWh) (incl. imports)

	2018-19	2019-20	2020-21
Indian Grid (Connected System)	0.9603	0.9555	0.9405

Weighted Generation Operating Margin	
Indian Grid (Connected System)	0.9522

Step 5: Calculate the build margin (BM) emission factor ($EF_{grid,BM,y}$)

As per CDM Methodological tool "Tool to calculate the emission factor for an electricity system" (Version 07.0, EB 100, Annex 4) para 72:

In terms of vintage of data, project owner can choose between one of the following two options:

- a) **Option 1-** for the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of PSF submission to the GCC verifier for project verification. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the GCC Verifier. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period.
- b) **Option 2-** For the first crediting period, the build margin emission factor shall be updated annually, ex post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex-ante as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

Note: In GCC there is fixed crediting period for 10 years, no renewal (2nd or 3rd) of crediting period will be there. So, BM is fixed for the entire crediting period.

Option 1 as described above is chosen by PP to calculate the build margin emission factor for the project activity. BM is calculated ex-ante based on the most recent information available at the time of submission of PSF and is fixed for the entire crediting period.

Build Margin (tCO ₂ /MWh) (not adjusted for imports)	
Year	2020-21
Indian Grid	0.8653

Step 6: Calculate the combined margin (CM) emission factor ($EF_{grid,CM,y}$)

As per CDM Methodological tool "Tool to calculate the emission factor for an electricity system" (Version 07.0, EB 100, Annex 4) para 81: -

The calculation of the combined margin (CM) emission factor ($EF_{grid,CM,y}$) is based on one of the following methods:

- a) Weighted average CM: or
- b) Simplified CM

PP has chosen option (a) i.e., weighted average CM to calculate the combined margin emission factor for the project activity.

The combined margin emissions factor is calculated as follows:

$$EF_{grid,CM,y} = EF_{grid,OM,y} * W_{OM} + EF_{grid,BM,y} * W_{BM}$$

Where;

$EF_{grid,CM,y}$	Build margin CO ₂ emission factor in year y (t CO ₂ / MWh)
$EF_{grid,OM,y}$	Operating margin CO ₂ emission factor in year y (t CO ₂ /MWh)
W_{OM}	Weighting of operating margin emissions factor (per cent)
W_{BM}	Weighting of build margin emissions factor (per cent)

The following default values should be used for W_{OM} and W_{BM} as per Para 86 of Tool 7:

a) Wind and solar power generation project activities: $w_{OM} = 0.75$ and $w_{BM} = 0.25$ (owing to their intermittent and non-dispatchable nature) for the first crediting period and for subsequent crediting periods;

b) All other projects: $w_{OM} = 0.5$ and $w_{BM} = 0.5$ for the first crediting period, and $w_{OM} = 0.25$ and $w_{BM} = 0.75$ for the second and third crediting period,8 unless otherwise specified in the approved methodology which refers to this tool.

For Hydro project activities $W_{OM} = 0.5$ and $W_{BM} = 0.5$ (owing to their intermittent and non-dispatchable nature) for the second crediting period and for subsequent crediting periods. Since project activity is of power generation by using Hydro energy, the above weightage has been considered for OM and BM.

Therefore,

$$EF_{grid,CM,y \text{ India}} = 0.9522 * 0.5 + 0.8653 * 0.5 \\ = 0.9087 \text{ CO}_2/\text{MWh}$$

- Combined Margin Emission Factor for Nepal:

$$EF_{grid,CM,y \text{ Nepal}} = 0.0 \text{ CO}_2/\text{MWh}$$

Nepal Electricity Generation ³	6052000	MWh
Power Purchase from India ⁴	2826000	MWh

- Connected system weighted average combined Emission Factor: =

$$\frac{\text{Nepal Generation} * EF_{grid,CM,y, \text{ Nepal}} + \text{Power purchase from India} * EF_{grid,CM, y, \text{India}}}{\text{Nepal Electricity Generation} + \text{Power Purchase from India}}$$

Connected system weighted average combined Emission Factor = 0.3031 tCO₂/MWh

³ https://www.nea.org.np/annual_report

⁴ https://cea.nic.in/wp-content/uploads/tpe_cc/2022/02/User_Guide_ver_17_2021.pdf

5.2 Project Emissions

As per the approved consolidated Methodology AMS I.D Version 18.0 para 24 “For most renewable energy power generation project activities, $PE_y = 0$. However, some project activities may involve project emissions that can be significant. These emissions shall be accounted for as project emissions by using the following equation:

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y}$$

Where:

PE_y = Project emissions in year y (t CO₂e/yr)

$PE_{FF,y}$ = Project emissions from fossil fuel consumption in year y (t CO₂/yr)

$PE_{GP,y}$ = Project emissions from the operation of dry, flash steam or binary geothermal power plants in year y (t CO₂e/yr)

$PE_{HP,y}$ = Project emissions from water reservoirs of hydro power plants in year y (t CO₂e/yr)

As the project activity is the installation of a new grid-connected hydropower plant and does not involve any project emissions from fossil fuel, operation of dry, flash steam or binary geothermal power plants, and from water reservoirs of hydro power plants. Therefore $PE_{FF,y}$, $PE_{GP,y}$, $PE_{HP,y}$ are equal to zero and thus, $PE_y = 0$.

5.3 Leakage

Since the project activity instances to be included into the grouped project activity are renewable energy projects based on hydropower, no leakage emissions are considered - in line with paragraph 27 of the small-scale methodology AMS.I.D (version 18.0).

5.4 Estimated Net GHG Emission Reductions and Removals

As per the para 28 of AMS-I.D version 18.0, the formula to calculate the emission reduction is

$$ER_y = BE_y - PE_y - LE_y$$

As the project activity is a hydropower project, there is no any leakage emissions from the project activity.

Hence, $LE_y = 0$

Therefore, Emission Reductions for this project activity are calculated as follows:

$$ER_y = BE_y - PE_y$$

Where:

ER_y = Emission reductions in year y (t CO₂e/yr)

BE_y = Baseline emissions in year y (t CO₂e/yr)

PE_y = Project emissions in year y (t CO₂e/yr)

Therefore, Net GHG Emission Reductions and Removals are calculated as follows:

$$ER_y = BE_y - PE_y$$

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
<u>01-August-2021-- 31-July-2022)</u>	22,737	0	0	22,737
<u>01-August-2022-- 31-July-2023)</u>	22,737	0	0	22,737
<u>01-August-2023-- 31-July-2024)</u>	22,737	0	0	22,737
<u>01-August-2024-- 31-July-2025)</u>	22,737	0	0	22,737
<u>01-August-2025-- 31-July-2026)</u>	22,737	0	0	22,737
<u>01-August-2026-- 31-July-2027)</u>	22,737	0	0	22,737
<u>01-August-2027-- 31-July-2028)</u>	22,737	0	0	22,737
Total	159,162	0	0	159,162

6 MONITORING

6.1 Data and Parameters Available at Validation

Data / Parameter	EF_{grid, OM,y} (Indian Grid)
Data unit	tCO ₂ e/MWh
Description	Operating Margin CO ₂ emission factor in year y
Source of data	CO ₂ Emission Database, Version 17.0, October- 2021 published by Central Electricity Authority (CEA), Government of India.
Value applied:	0.9522
Justification of choice of data or description of measurement methods and procedures applied	Calculated as the last 3-year (2018-19, 2019-20, 2020-21) generation-weighted average, sourced from Baseline CO ₂ Emission Database, Version 17.0, October- 2021 published by Central Electricity Authority (CEA), Government of India.
Purpose of Data	For the calculation of the Baseline Emission
Comments	This parameter is fixed ex-ante for the entire crediting period.

Data / Parameter	EF_{grid, BM,y} (Indian Grid)
Data unit	tCO ₂ e/MWh
Description	Build Margin CO ₂ emission factor in year y
Source of data	CO ₂ Emission Database, Version 17.0, October- 2021 published by Central Electricity Authority (CEA), Government of India.
Value applied:	0.8653
Justification of choice of data or description of measurement methods and procedures applied	Calculated as the last year (2020-21) build margin, sourced from Baseline CO ₂ Emission Database, Version 17.0, October- 2021 published by Central Electricity Authority (CEA), Government of India.

Purpose of Data	For the calculation of the Baseline Emission
Comments	This parameter is fixed ex-ante for the entire crediting period.

Data / Parameter	EF_{grid, CM,y} (Indian Grid)
Data unit	tCO ₂ e/MWh
Description	Combined Margin CO ₂ emission factor in year y
Source of data	CO ₂ Emission Database, Version 17.0, October- 2021 published by Central Electricity Authority (CEA), Government of India.
Value applied:	0.9087
Justification of choice of data or description of measurement methods and procedures applied	The combined margin emissions factor is calculated as follows: $EF_{grid,CM,y} = EF_{grid,OM,y} * W_{OM} + EF_{grid,BM,y} * W_{BM}$ Where: EF_{grid,BM,y}= Build margin CO₂ emission factor in year y (tCO₂/MWh) EF_{grid,OM,y}= Operating margin CO₂ emission factor in year y (tCO₂/MWh) W_{OM} = Weighting of operating margin emissions factor (%) = 75% W_{BM}= Weighting of build margin emissions factor (%) = 25%
Purpose of Data	For the calculation of the Baseline Emission
Comments	This parameter is fixed ex-ante for the entire crediting period.

Data / Parameter	EF_{grid, CM,y} (Nepal Grid)
Data unit	tCO ₂ e/MWh
Description	Combined Margin CO ₂ emission factor in year y
Source of data	Harmonized IFI Default Grid Factors 2021 v3.2
Value applied:	0.0

Justification of choice of data or description of measurement methods and procedures applied	NA
Purpose of Data	For the calculation of the Baseline Emission
Comments	This parameter is fixed ex-ante for the entire crediting period.

6.2 Data and Parameters Monitored

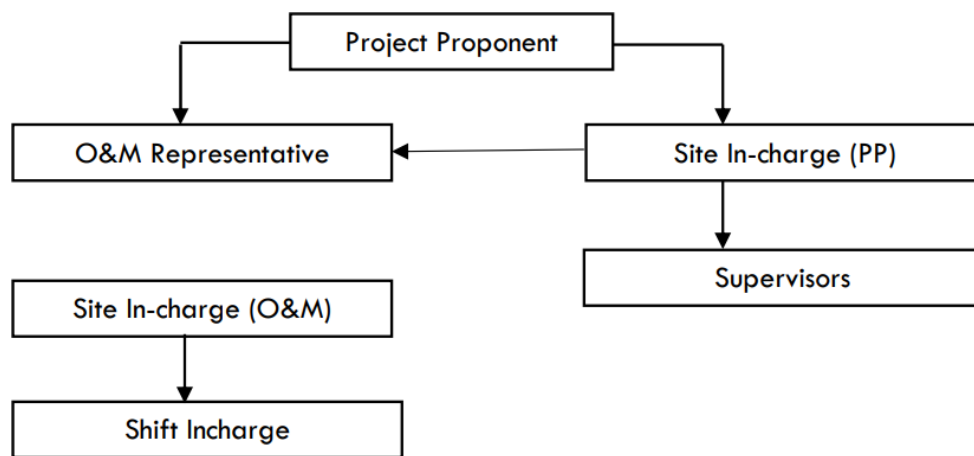
Data / Parameter	EG_P, y
Data unit	MWh/y
Description	Quantity of Net electricity generation supplied by the project plant/unit to the grid in year y
Source of data	Credit note/ JMR/B Form reports/ monthly generation report from state electricity board/DISCOM
Description of measurement methods and procedures applied	-
Frequency of monitoring/recording	Continuous measurement & monthly recording
Value applied:	75,016
Monitoring equipment	Will be provided on later stage of Verification
QA/QC procedures applied	Meters will be calibrated once in 5 years
Purpose of data	Calculation of baseline emissions
Calculation method	The Joint Meter Readings (JMR) taken every month from each meter are added up to arrive at the net value of electricity generated by project activity. Also bills of net metering will be considered for determination of net electricity generated by project activity.
Comments	-

6.3 Monitoring Plan

The monitoring plan is developed in accordance with the modalities and procedures for CDM project activities. The monitoring plan, which will be implemented by the project participant describes about the monitoring organisation, parameters to be monitored, monitoring practices, quality assurance, quality control procedures, data storage and archiving.

Organizational structure and responsibilities:

The authority and responsibility for registration, monitoring, measurement, reporting and reviewing of the data rests with the project participant. PP proposed the following structure for data monitoring, collection, data archiving and calibration of equipment for this project activity. The team comprises of the following members:



Responsibilities of Site Incharge (PP): Overall functioning and maintenance of the project activity, the Site incharge shall coordinate with the O&M operator as well as the site supervisors.

Responsibilities of O&M Representative: Co-ordination between Site incharge of the O&M operator as well as the project participant and further report to PP head office.

Responsibilities of Site In-charge (O&M Operator): Responsibility for maintaining the data records, ensures completeness of data, and reliability of data (calibration of equipment) as well as data recording for all the parameters.

Responsibilities of Shift In-charge: Responsibility for day-to-day data collection and maintains day to day monitored data.

Data Measurement

The export and import energy or net electricity displaced by project will be measured continuously using above mentioned energy meter located after inverters. Readings of meters shall be taken on monthly basis by authorized officer of developer in the presence of PP or representative of PP. In the event of failure of generation meter, the user energy meter or inverter generation value will be used in monitoring the electricity generation data.

Data collection and archiving

Readings from meters will be collected in the presence of the plant in-charge. Export and import or net electricity displaced by project would be recorded and stored in logs as well as in electronic form on a daily basis. The records are checked periodically by the Plant Manager and discussed thoroughly with the plant supervisor. The period of storage of the monitored data will be 2 years after the end of crediting period or till the last issuance of VERs for the project activity whichever occurs later.

Emergency preparedness

The project activity will not result in any unidentified activity that can result in substantial emissions from the project activity. No need for emergency preparedness in data monitoring is visualized.

QA/QC Procedures

There energy meters are installed at the respective project site. The calibration of all the meters will be undertaken at required intervals (once is five years as per CEA notification) and faulty meters will be duly replaced immediately. Records of calibration certificates will be maintained for verification purposes. Hence, a reliable method will be ensured with monitoring of the parameters. The invoice records will be used and kept for cross checking the consistency of the recorded data.

Sampling Approaches

Not required

Personnel training

In order to ensure a proper functioning of the project activity and a properly monitoring of emission reductions, the staff will be trained. The plant helpers will be trained in equipment operation, data recording, reports writing, operation and maintenance and emergency procedures in compliance with the monitoring plan.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

Data / Parameter	EG _{PJ,y}
Data unit	MWh/y
Description	Quantity of Net electricity generation supplied by the project plant/unit to the grid in year y
Value applied:	-
Comments	-

7.2 Baseline Emissions

Baseline emission is the product of Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity multiplied by the Combined margin CO₂ emission factor for grid connected power generation

$$BE_y = EG_{PJ,y} \times EF_{grid,y}$$

Where,

BE_y = Baseline Emissions in year y; t CO₂

EG_{PJ,y} = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)

EF_{grid, y} = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system Version 7.0” (t CO₂/MWh).

7.3 Project Emissions

As per methodology AMS I.D Version 18.0, for renewable energy projects, there is no any project emissions occurred.

Hence, PE_y= 0

7.4 Leakage

As per methodology AMS I.D Version 18.0, for renewable energy projects, there is no any leakage emissions occurred.

Hence, LE_y= 0

7.5 Net GHG Emission Reductions and Removals

This section will complete during validation and verification stage.

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
Year A (DD-Month-YYYY-- DD-Month-YYYY)				
Year...				
Total				

This section will complete during validation and verification stage.

<u>Ex-ante emissions reductions /removals</u>	<u>Achieved emissions reductions /removals</u>	<u>Percent difference</u>	<u>Justification for the difference</u>